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SeS-CLIP: A CLIP-based model for semi-supervised video anomaly detection ☆

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ABSTRACT

Video Anomaly Detection (VAD) is a pivotal research domain with numerous real-world applications. Given the scarcity of annotated anomalous events, semi-supervised anomaly detection approaches primarily rely on training with normal samples, identifying deviations during inference as anomalies. Recent advancements in Contrastive Language-Image Pretraining (CLIP) models have demonstrated remarkable performance in vision tasks, showcasing robust capabilities in representation learning. However, current CLIP-based VAD models often adopt weakly supervised training paradigms requiring video-level annotations, thus constraining their broader applicability in VAD. In this study, we propose an innovative CLIP-based semi-supervised model for VAD. This work is the first to exploit the Large Vision-Language Model's ability to enhance the interpretation of complex visual scenes, facilitating a focus on high-level semantic feature learning for semi-supervised VAD. By bridging visual and text modalities, our model maps low-level visual representations to high-level semantic representations and effectively models their distribution for anomaly detection. Furthermore, we introduce synthetic anomalous text samples into semi-supervised VAD tasks to enhance feature divergence. Extensive experiments demonstrate that our model achieves SOTA performance on key semi-supervised VAD benchmark datasets.

1. Introduction

Video anomaly detection (VAD) aims to identify instances that deviate significantly from established normal patterns in video footage [1], such as traffic accidents, violent incidents, and unauthorized intrusions, which are critical for maintaining public safety in areas with widespread surveillance. However, detecting these anomalies is challenging due to their rarity and diversity, which complicates the development of accurate detection systems. VAD methods are classified into five types based on the kind of training datasets: unsupervised, semi-supervised, weakly supervised, fully supervised, and Open-Set. Among them, the fully supervised method is equipped with accurate anomaly labeling for each anomalous sample, while the weakly supervised method has coarse video-level anomaly labels. Due to the inclusion of anomalous samples in training, models typically achieve high accuracy in VAD [2].

One of the primary challenges in VAD is the difficulty of acquiring comprehensive labeled datasets for supervised learning. Anomalous events are not only infrequent but also diverse, making manual annotation a time-consuming and resource-intensive process. This limitation restricts the application of both weakly supervised and fully supervised VAD models. As a result, researchers are increasingly exploring a semi-supervised learning approach. This technique allows models to learn exclusively from normal instances and identify deviations without exhaustive labeling [3]. Compared with fully and weakly supervised methods, semi-supervised learning eliminates the need to collect rare anomalous samples, thereby positioning it as a more versatile and widely applicable framework for VAD with significant research potential.

Numerous semi-supervised VAD methods have been proposed, with generative models such as Future Frame Prediction (FPM) [4], Generative Adversarial Networks (GAN) [5], and Diffusion Models (DM) [6] gaining popularity. These models are effective because anomalous samples unseen during training are generally more difficult to reconstruct or predict than normal samples. However, semi-supervised VAD models face two main challenges: (1) Generative models tend to "over-generalize to anomalies" [7], using low-level features, such as color and texture, to reconstruct anomalous samples, which results in more effective reconstruction of anomalies than expected. (2) The lack of anomalous samples during training makes it difficult to establish a precise decision boundary between normal and abnormal classes, resulting in suboptimal performance.

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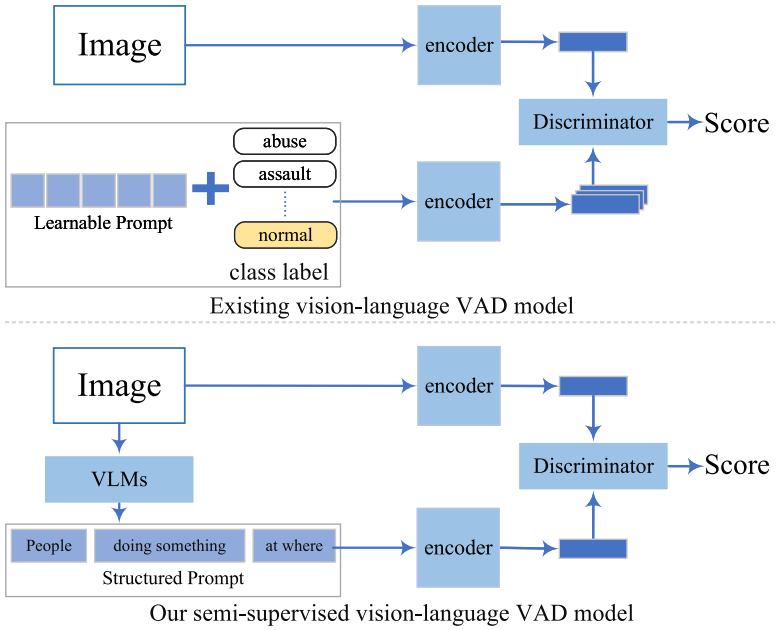


Fig. 1. Traditional CLIP-based models are limited to the weakly supervised VAD paradigm. By incorporating an LVLM to generate structured text, we extend these models to the semi-supervised VAD paradigm.

Using CLIP [8] to address "over-generalizing to anomalies" has become increasingly popular in recent VAD research [9], owing to its exceptional performance in various vision-language tasks. By learning associations between vision and language features through pre-training on large-scale datasets, CLIP's vision encoder emphasizes high-level semantic features, effectively mitigating issues caused by reliance on low-level features. However, as illustrated in Fig. 1, existing CLIP-based anomaly detection frameworks require aligning anomalous text prompts with corresponding anomalous image features during training. Consequently, CLIP is primarily used within a weakly supervised training paradigm, which relies more heavily on annotations than a semi-supervised approach.

In this paper, we introduce a novel semi-supervised VAD framework called SeS-CLIP, which consists of two primary modules: a text generation module and a feature contrast module. The text generation module employs a Large Vision-Language Model (LVLM) to generate descriptive text for each video frame, facilitating anomaly detection. Subsequently, the CLIP-based feature contrast module calculates the similarity between image and text embeddings to distinguish anomalies. Unlike conventional CLIP-based models, our approach eliminates the need to align abnormal image features with corresponding text, thereby precluding the inclusion of anomalous images in the training set and facilitating a semi-supervised training paradigm. Our method aligns image-text features exclusively for normal samples during training, identifying anomalies as samples with low image-text feature similarity during inference. Additionally, to address the challenge of the absence of anomalous training samples, we incorporate synthetic anomalous texts during the training phase. By utilizing contrastive learning, we enhance the distinction between normal and anomalous text features, ultimately improving the accuracy of anomaly detection.

Our main contributions can be summarized as follows:

- We are the first to leverage the LVLM to enable semi-supervised VAD to effectively focus on learning high-level semantic features of video frames, thereby enhancing the interpretation of complex visual scenes.
- We propose an innovative semi-supervised VAD approach that utilizes the LVLM to bridge visual and text modalities by transferring low-level visual features to high-level semantic contexts. This method thoroughly learns the distribution of high-level semantics for effective anomaly detection.
- We introduce a synthetic anomaly text generation framework that extends semi-supervised VAD beyond the visual modality. Compared to visual data, synthetic anomalous text can better approximate the semantics of real anomalous events.
- Extensive experiments across three public VAD datasets validate the effectiveness of our framework, showing that the proposed framework outperforms existing state-of-the-art methods.

2. Related work

2.1. Semi-supervised VAD

Anomaly detection in computer vision can be classified into weakly supervised, semi-supervised, fully supervised, and unsupervised approaches, depending on the specific scenarios and variations in training datasets [2]. Our method falls under the semi-supervised anomaly detection category, which has seen significant advancements in recent years.

Semi-supervised anomaly detection methods are generally divided into generative and discriminative models. Generative methods have gained popularity, leveraging neural networks to either reconstruct current frames or predict future frames. Anomalies are identified by comparing discrepancies between generated and original frames. Liu et al. [4] introduced a future-frame prediction framework that uses prediction errors to identify anomalous events. Gong et al. [10] introduced a memory module that intentionally reduces reconstruction quality for anomalous samples. Hirschorn et al. [11] utilized human keypoint detection to generate skeletal data, thereby minimizing background noise effects. Yan et al. [12] enhanced the quality of generative frames using a diffusion model.

Discriminative VAD models [13], on the other hand, focus on characterizing statistical distributions of normal instances and establishing decision boundaries around them. While these models avoid accuracy degradation from inadequate generation capabilities, they encounter challenges in precisely identifying anomalies.

2.2. Pseudo anomalies

In semi-supervised frameworks, the absence of truly anomalous samples during training poses a challenge. To address this, some studies

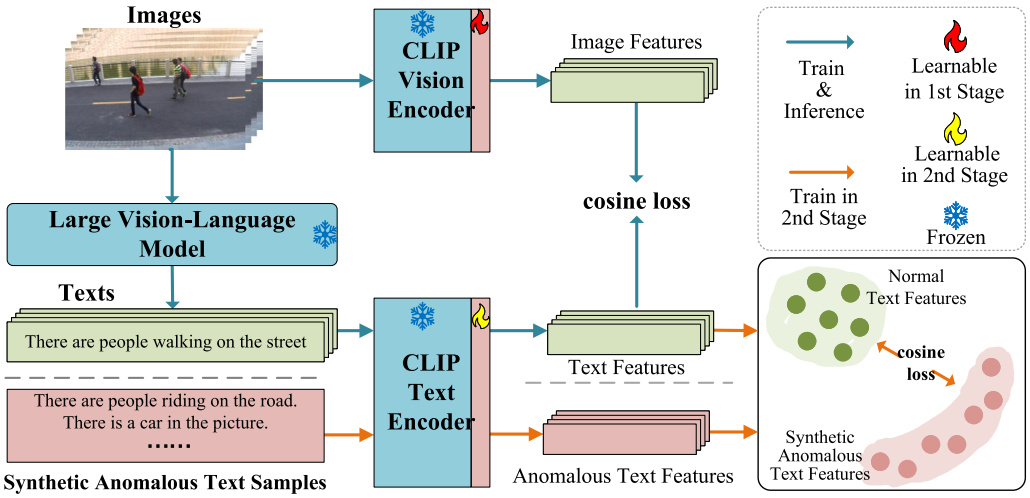


Fig. 2. Overview of the proposed SeS-CLIP framework. The LVLM first generates structured textual descriptions for video frames, which are then encoded together with the corresponding images by the CLIP-based feature contrast module. Training is performed in two stages: the first stage fine-tunes the vision encoder to align normal image–text pairs, while the second stage fine-tunes the text encoder using synthetic anomalous captions to enlarge the separation between normal and anomalous representations. During inference, the angular distance between image and text embeddings is used as the anomaly score.

generate pseudo-anomalies to enhance the decision boundary between normal and anomalous samples, thereby improving detection accuracy. Early methods typically involved perturbing normal video samples by adding noise, shuffling frame sequences, inserting extra patches, or integrating random images [14–17]. These approaches, however, result in pseudo-anomalies that significantly differ from real anomalies.

Recently, Ristea et al. [18] and Liu et al. [19] incorporated anomalous samples from the UBnormal dataset [20] during training. Wu et al. [21] employed AI-generated content (AIGC) models to create pseudo-anomalous short videos tailored to specific anomaly types across various scenarios. While these methods approximate real anomalous samples more closely, they still have limitations. Synthetic anomalous images often diverge from the original videos in aspects such as bitrate, texture quality, and color space, complicating the estimation of whether the model might learn misleading anomalies from these unexpected factors. Additionally, datasets have varying definitions of anomalies, and it may not always be feasible to find a public dataset with compatible definitions for all scenarios.

In contrast, our approach first utilizes an LVLM to generate structured text descriptions, followed by the creation of synthetic anomaly samples based on the textual structure. This transformation from images to text effectively mitigates the aforementioned issues.

2.3. Vision-language models in VAD

In recent years, there has been a surge in CLIP-based VAD models [22]. Unlike traditional methods that solely rely on visual features, CLIP-based approaches leverage both visual and textual data to establish more robust classification boundaries within the semantic space. This dual-modality capability enables better adaptation to dynamic scene changes [3]. However, many of these models incorporate a crucial training step: utilizing learnable prompt techniques to align embeddings of anomalous frames with corresponding anomaly-type text prompts. Consequently, these models depend on weakly supervised VAD datasets with clear and limited anomaly types, a requirement that significantly restricts the versatility and generalizability of CLIP models in the VAD domain.

Alongside CLIP-based VAD models, LVLM-based VAD methods have also gained attention [23]. These methods benefit from rich semantic knowledge acquired during the pretraining phase, enabling them to focus on general anomalous events such as explosions and fights in datasets such as UCF-Crime [24] and XD-Violence [25]. Despite their strengths, they often struggle to detect artificially defined anomalies, such as skateboarding and running, as highlighted in the ShanghaiTech dataset [26].

Moreover, Yang et al. [27] proposed a denylist approach that involves generating frame descriptions with CogVLM-17B [28] and using GPT-4 to compile an extensive inventory of anomalous events. This method checks whether listed anomalies occur in the text associated with each frame, facilitating the detection of predefined anomalous events. However, this strategy is inherently limited to identifying anomalies within a pre-defined list, thereby failing to capture the full spectrum of rare and diverse anomalous phenomena that might occur outside the enumerated scenarios.

3. Method

Our primary motivation is to develop a CLIP-based model that effectively learns high-level semantic features from text descriptions of normal video frames, specifically for semi-supervised video anomaly detection. During inference, the optimized model yields significantly higher image-text feature similarities for normal samples than for anomalous ones.

As shown in Fig. 2, our framework comprises two main modules: the structured text generation module and the feature contrast module. The structured text generation module employs an LVLM to create structured textual descriptions for each video frame. These descriptions, along with the corresponding images, are processed through CLIP encoders within the feature contrast module to obtain feature vectors. The angular distance between these feature vectors serves as the anomaly scoring metric. During training, synthetic anomalous samples are generated, and contrastive learning is applied to enhance the feature-representation differences between normal and anomalous text. This design enhances the separation between normal and anomalous representations for VAD.

3.1. Problem formulation

Our objective is to generate image and text features from consecutive video frames and their corresponding textual descriptions, and then estimate whether these features belong to the learned distribution. Specifically, given a video sequence $X = \{x_1, x_2, \ldots, x_k\}$ comprising k consecutive frames, the LVLM first produces the corresponding text set $T = \{t_1, t_2, \ldots, t_k\}$. We denote the image and text embeddings of the nth frame by v_n^x and v_n^t, respectively. The anomaly score $Score_n$ is calculated by the angular distance between v_n^x and v_n^t.

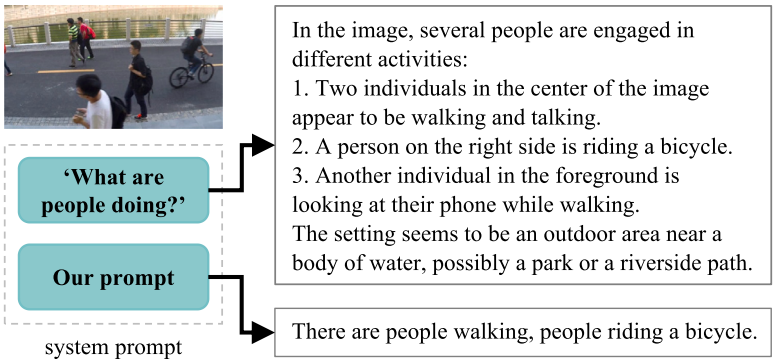


Fig. 3. Comparison of two types of prompts. The prompt we proposed effectively compresses the text length while preserving the relevant information.

Table 1
Statistics of output text lengths for the ShanghaiTech dataset under different prompts.

	Average words	Outputs exceeding 77 tokens (%)
Prompt from [27]	65.36	97.02
Our prompt	9.24	0

3.2. Structured text generation module

The structured text generation module is the first component of SeS-CLIP. This module leverages a frozen LVLM to transform video frames into structured textual descriptions. The structured format facilitates the controlled construction of synthetic anomalous captions during training. The structured text generation process can be represented by the following equation:

$$t_n = f(x_n, p) \tag{1}$$

where f denotes the frozen LVLM and p denotes the input prompt.

In the work by Yang et al. [27], the prompt "What are people doing? What is in the images other than people?" was used by the LVLM to generate descriptions of people's behavior and other objects. As illustrated in Fig. 3 and Table 1, this approach often results in unstructured and excessively long descriptions, which not only increases the time required to generate text for each frame but also exceeds the input text length limit of 77 tokens for the pretrained CLIP model, thereby affecting the accuracy of video anomaly detection.

To overcome these limitations, we have devised a more sophisticated prompt. For action anomalies, we use the following prompt: "*Whether there are people in the picture: Summarize what the people in the picture are doing, group people performing similar actions together; do not describe them separately. Focus on full-body movements and ignore localized movements. If people are present in the image, the fixed output format is: 'There are people [doing action X], people [doing action Y].'; If no people are present, then the fixed output is 'There are no people in the picture'.*" This prompt is designed to guide the LVLM in sequentially analyzing the behavioral details of individuals in the scene, thereby improving the completeness and accuracy of the generated descriptions.

Additionally, we incorporate several other text prompts in the form: "*Whether there are (A) in the picture: If (A) are present, add a sentence: There are (A) in the picture. but if no (A) are present, do not add any sentences.*" to facilitate the detection of other anomaly types. Here, *A* typically stands for "*other anomalous events*" and can be extended with additional sentences tailored to the known anomaly categories in the given scene, thereby facilitating the detection of various anomaly types.

3.3. Feature contrast module

We developed the feature contrast module as the second component of the SeS-CLIP framework for video anomaly detection. In this module, CLIP encoders are used to extract embedding features from both images and text, with the text being the output from the structured text generation module. The architecture of the feature contrast module is outlined as follows:

$$v_n^x = g^i(x_n) \tag{2}$$

$$v_n^t = g^t(t_n) \tag{3}$$

$$Score_n = \arccos\left((v_n^x \cdot v_n^t)/(\|v_n^x\| \, \|v_n^t\|)\right) \tag{4}$$

where g^i and g^t denote the vision encoder and text encoder of the CLIP model, respectively. We utilize the angular distance between v_n^x and v_n^t as the anomaly score $Score_n$. Samples with higher anomaly scores are identified as anomalous, allowing for effective anomaly detection based on the angular distance between image and text embeddings.

3.4. Training

The training objectives are twofold: minimizing $Score_n$ for normal samples and maximizing it for anomalous samples. To achieve these goals, we structure the training process into two distinct phases. In the first phase, the vision encoder is trained to accomplish the first objective. In the second phase, synthetic anomalous text samples are generated, and contrastive learning is employed to train the text encoder, thereby achieving the second objective. To prevent representation collapse, it is essential to train the vision encoder and text encoder in separate phases.

3.4.1. Vision encoder training

The initial training stage focuses on minimizing $Score_n$ in normal samples. Due to the semantic differences between the task-specific captions generated by the LVLM for anomaly detection and the text in the CLIP pre-training dataset, fine-tuning the vision encoder is necessary to adapt pre-trained features to the target domain. This stage uses a training approach similar to traditional semi-supervised learning: the model is trained to produce smaller angular distances for normal image–caption pairs than those expected for anomalous pairs.

For each batch during the first training phase, we freeze most of the model's parameters and fine-tune only the last two Transformer layers of the vision encoder. To minimize the angular distance anomaly score of normal samples, the objective function for this phase is represented by:

$$Score_{(n,m)} = \arccos\left((v_n^x \cdot v_m^t)/(\|v_n^x\| \, \|v_m^t\|)\right), \quad m \in [n-3, n+3] \cap \mathbb{N} \tag{5}$$

$$Score'_n = \frac{1}{3}\sum_{r=1}^{3} Score_{n,(r)}, \quad Score_{n,(1)} \le Score_{n,(2)} \le \cdots \tag{6}$$

$$L^{v1} = \frac{1}{\pi b}\sum_{n=1}^{b} Score'_n \tag{7}$$

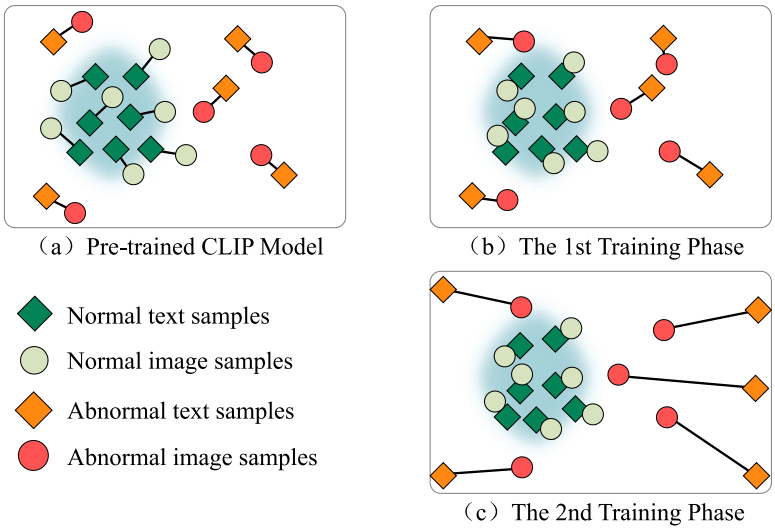


Fig. 4. Schematic diagram of image and text features during two-stage training.

where b is the batch size for L^{v1}. $Score'_n$ refers to the mean of the three smallest angular distances for the nth frame, computed between the image embedding v_n^x and the caption embeddings v_m^t from adjacent frames. According to the statistical data in Section 4.5 and Fig. 9, LVLMs occasionally misidentify information in images, such as misrecognizing walking as skateboarding or throwing a frisbee. Based on temporal continuity, captions of adjacent frames are generally expected to be semantically consistent. Therefore, we use $Score'_n$ rather than the self-matching loss $Score_n$ to prevent the model from learning from incorrect text features.

To mitigate catastrophic forgetting on out-of-distribution (OOD) samples, we employ a frozen CLIP vision encoder as the teacher model and minimize the mean absolute difference between the visual embeddings produced by the trainable and frozen encoders. Here, g^{if} and x_n^c refer to the frozen vision encoder and the images from the pre-training dataset, and L^v refers to the loss function for vision encoder training.

$$v_n^c = g^i(x_n^c) \tag{8}$$

$$v_n^{cf} = g^{if}(x_n^c) \tag{9}$$

$$L^{v2} = \frac{1}{b}\sum_{n=1}^{b} \| v_n^c - v_n^{cf} \|_1 \tag{10}$$

$$L^v = L^{v1} + \alpha L^{v2} \tag{11}$$

3.4.2. Text encoder training

The second training stage focuses on maximizing $Score_n$ for anomalous samples. To achieve this, we train the text encoder using synthetic anomalous text samples, aiming to enhance the difference between the feature representations of normal texts and those of synthetic anomalies.

During the initial training stage, we minimized the feature representation differences between images and texts for normal samples. However, this traditional semi-supervised approach may lead to an uncertain distribution of feature representations for anomalous image samples.

To address this, as illustrated in Fig. 4, we maximize the feature representation differences between normal and synthetic anomalous text samples. This separation is expected to produce higher anomaly scores for anomalous samples at inference, thereby strengthening the model's ability to distinguish between normal and anomalous instances.

In Section 3.2, we employ an LVLM to generate structured textual descriptions, facilitating the creation of more realistic synthetic anomalous samples. The structured text is formatted as follows: *"There are people [performing action] [at location]. There are ['anomalous event' or 'no anomalous event'] in the picture"*. Using this format, we derive a list of anomalous events from the dataset's specified anomaly types and a list of locations from common directional terms observed in the training set. By randomly combining anomalous events from a predefined list with these common locations, we effectively synthesize pseudo-anomalous text samples. This combination method may result in actions that do not match the location, such as *'There are people riding a bicycle on a ledge'*. Although random combinations may occasionally produce semantically implausible captions, these captions are used only as negative samples. This design encourages separation from the normal-text distribution rather than requiring the synthetic captions to represent realistic anomalous events precisely.

In the semi-supervised video anomaly detection task, anomaly categories are open-ended and cannot be exhaustively enumerated. Similar to the purpose of generating anomaly samples in [18], our use of synthetic anomaly samples is for generating negative samples for contrastive learning, rather than enumerating all anomaly types. Therefore, the model should be able to detect anomalies not explicitly listed. This capability is experimentally validated in Section 4.4.

During each batch in the second training phase, the majority of the model's parameters are frozen, and only the last two Transformer layers of the text encoder are fine-tuned. We generate c synthetic anomalous captions in each mini-batch, maximizing the divergence between their feature representations and those of the training set. The objective function is defined as follows:

$$S^t_{(i,j)} = (v_i^t \cdot v_j^{st}) / (\| v_i^t \| \ \| v_j^{st} \|) \tag{12}$$

$$L = \frac{1}{bc}\sum_{i=1}^{b}\sum_{j=1}^{c} S^t_{(i,j)} \tag{13}$$

where v_j^{st} refers to the embeddings of the synthetic samples, and L represents the average cosine similarity between v_j^{st} and v_i^t. To prevent an increase in $Score_n$ for normal samples, L^{v1} remains necessary. The final objective function of the second training stage, balanced by β, is designed as follows:

$$L^t = \beta L^{v1} + (1-\beta)L \tag{14}$$

Table 2
AUC performance of various VAD methods. PA indicates the use of pseudo-anomalies. Ped2, SHT, and UB refer to the UCSD-Ped2, ShanghaiTech, and UBnormal datasets, respectively.

Method	Venue	PA	Ped2	SHT	UB
MemAE [10]	ICCV-19	–	94.1%	71.2%	–
CDAE [30]	ECCV-20	–	96.5%	73.3%	–
RUVAD [31]	TNNLS-21	–	96.3%	76.6%	–
STDAD [32]	PR 2022	–	96.7%	73.7%	–
STJP [33]	ECCV-22	–	99.0%	84.3%	–
MAAM-Net [34]	PR 2023	–	97.7%	–	71.3%
FPDM [12]	ICCV-23	–	–	78.6%	62.7%
STG-NF [11]	ICCV-23	–	–	85.9%	71.8%
MoCoDAD [35]	ICCV-23	–	–	77.6%	68.3%
SSPCAB [36]	TPAMI-23	–	–	83.7%	–
MULDE [13]	CVPR-24	–	–	81.3%	72.8%
TrajREC [37]	WACV-24	–	–	77.9%	68.0%
AnomalyRuler [27]	ECCV-24	–	97.9%	85.2%	71.9%
TIPNC [38]	PR 2025	–	98.6%	83.7%	–
ADSM [39]	ICCV-25	–	–	84.5%	–
GiCiSAD [40]	WACV-25	–	–	78.0%	68.6%
SC-MAFC [41]	PR 2026	–	**99.4%**	84.6%	–
LNTR [16]	CVPR-21	✓	96.5%	75.9%	–
BAF [17]	TPAMI-23	✓	98.7%	82.7%	59.3%
PFMF [19]	CVPR-23	✓	–	85.0%	–
SDMAE [18]	CVPR-24	✓	98.4%	79.1%	58.5%
OVVAD [42]	CVPR-24	✓	–	–	62.9%
SeS-CLIP(ours)	–	✓	96.9%	**87.0%**	**73.4%**

4. Experiments and discussion

4.1. Datasets

Empirical evaluations were conducted on three widely used video anomaly detection (VAD) datasets: UCSD Ped2 [29], ShanghaiTech [26], and UBnormal [20]. ShanghaiTech and UBnormal were the most commonly used test datasets for semi-supervised VAD models in recent years, whereas the Ped2 dataset was used to evaluate our model's performance in low-definition video scenes. The details are as follows:

- The UCSD Ped2 dataset contains 16 training videos (2.6k frames) and 12 test videos (2.0k frames), with a frame resolution of 360 × 240 px. It includes five anomalous events, such as bicycles, skateboards, and other objects intruding into the pedestrian area.
- The ShanghaiTech dataset is derived from natural university street scenes captured by ShanghaiTech University. The training set comprises 13 scenes, represented by 330 videos, while the test set comprises more than 40,000 frames stored in image format. Only pedestrians walking or standing normally were considered normal samples, whereas abnormal walking motions and vehicles such as cars and bicycles were considered anomalies.
- The UBnormal dataset is a VAD dataset generated with Cinema4D. It includes 29 scenes totaling more than 230,000 frames. In this dataset, people walking and vehicles moving normally on the road under any weather conditions are considered normal samples, while instances of fighting, falling, car crashes, fire, and smoke are deemed abnormal.

4.2. Implementation details

The structured text generation module employs the Qwen2.5-VL-7B-Instruct pretrained model. We utilize the COCO 2017 [43] training set as an auxiliary general-purpose image dataset for feature-preservation training. For the Ped2 and ShanghaiTech datasets, the input image size is retained at its original dimensions, with *max-new-tokens* set to 25 and *temperature* set to 0.05. For the UBnormal dataset, the input images are resized to 896 × 504, with `max_new_tokens` set to 35 and `temperature` set to 0.01. The feature contrast module utilizes the ViT-B/32 variant of the pretrained CLIP model, with input images resized to 224 × 224 and a maximum text length of 77 tokens. Training spans 40 epochs, with the first 16 epochs comprising the first phase and the remaining epochs comprising the second. We use a base learning rate of 2.1×10^{-5}, together with a cosine learning-rate scheduler. The hyperparameters α and β in the objective function are set to 1 and 0.65, with a batch size of 384 and 9 synthetic anomalous samples generated per batch. Following the benchmark source code from UBnormal, we apply Gaussian filtering to the anomaly scores for each video, with σ set to 20 for the ShanghaiTech and UCSD Ped2 datasets and 24 for the UBnormal dataset. All experiments are conducted on a single NVIDIA RTX A6000 Graphics card with a GA102 GPU and 48 GB of VRAM, coupled with an AMD Threadripper 5955WX CPU and 384 GB of RAM.

In the synthetic anomalous sample generation phase, based on the dataset information provided in previous studies [20,26,29], the anomalous event list for the UCSD Ped2 dataset includes *bikes, skaters, cars, and wheelchairs*. For the ShanghaiTech dataset, the list includes *bicycles, cars, cycling, skateboarding, running, playing, throwing objects, chasing, and fighting*. The UBnormal dataset considers anomalies such as *running, falling, fighting, sleeping, crawling, epileptic seizure, lying down, dancing, shuffling, injured walking, drunk walking, stumbling, car accidents, fire, smoke, jumping*.

4.3. Performance comparison

As illustrated in Table 2, we evaluated the performance of the proposed method on the aforementioned datasets. Our method was compared with 17 representative semi-supervised VAD models and 5 VAD models that utilize pseudo-anomalies. The results indicate that our approach achieved state-of-the-art performance on the ShanghaiTech and UBnormal datasets, underscoring its superior capability. The UCSD-Ped2 dataset, an early video anomaly detection benchmark, is characterized by single-scene settings, limited anomaly categories, and low image resolution, which has led to performance saturation, as early methods achieved near-perfect results. The Qwen2.5-VL-7B model occasionally overlooks objects at the image edges, such as cars and bicycles, which slightly affects the AUC metric, resulting in marginally lower performance compared to current state-of-the-art techniques.

4.4. Generalization test

This section evaluates the model's ability to detect anomaly types not present in the synthetic anomalous samples. Our objective is to demonstrate that our model does not solely depend on predefined anomaly lists and, unlike [27], does not require an exhaustive list of anomaly types to detect rare anomalies effectively. For clarity, the anomaly score presented here and in the subsequent section is scaled by a factor of $1/\pi$ times $Score_n$.

We conducted two experiments on video 2-161 from the ShanghaiTech test set, which depicts a campus street scene with bicycles as the predominant anomaly type. During training, we removed all bicycle-related terms from the list used to generate synthetic anomalous samples while keeping all other configurations unchanged. We then compared the performance of two models for this video.

In Fig. 5, the blue line represents our original model, while the green line represents the model trained without bicycle-related terms. Although removing these terms slightly reduces bicycle detection, a substantial disparity remains between the scores of anomalous and normal samples. The AUC in scene 2-161 decreases only marginally from 99.9% to 97.6%, indicating minimal impact. Additionally, as shown in Fig. 6, we manually generate images using AI-generated content (AIGC) techniques. These images, considered anomalies under the ShanghaiTech dataset's criteria but exceedingly rare, include people in flight (rare action anomalies), a Gundam mech (rare vehicle anomalies), and scenes of fire and ground fissures (disaster-type anomalies absent from the test set). The anomaly scores of these four images are

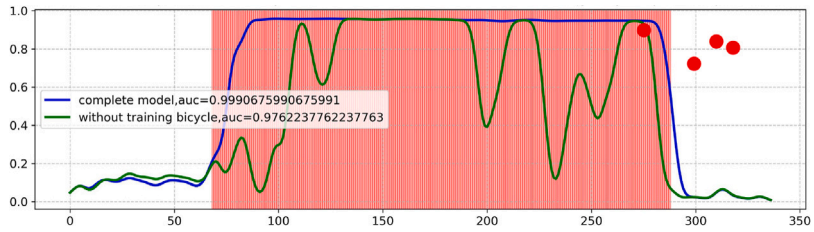


Fig. 5. The results for video 2-161 in ShanghaiTech. The blue line represents the results of the original training setting. The green line represents the results obtained when texts of bicycle are excluded during training. The red dots indicate anomaly instances of types that are not included in the original training set.

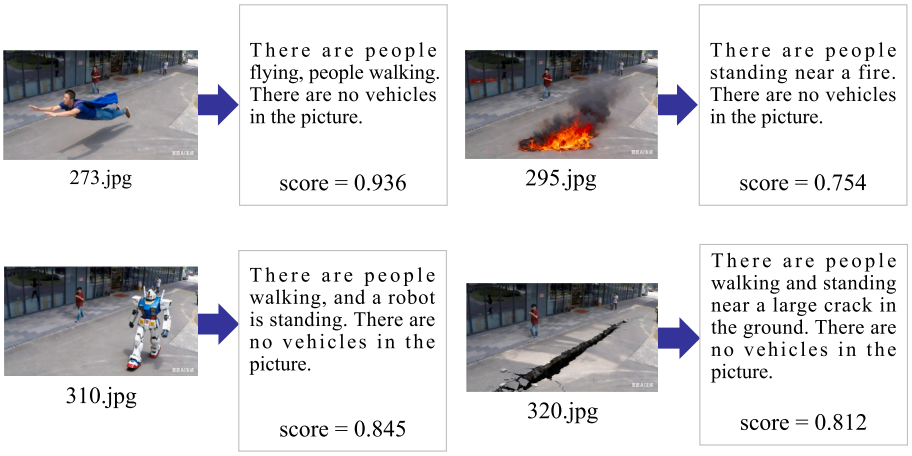


Fig. 6. Images generated by AIGC techniques and their corresponding results.

Table 3
AUC performance of our SeS-CLIP model and its variants. VE and TE denote vision encoder and text encoder fine-tuning, and ST refers to synthetic anomalous text.

VE	TE	ST	Remark	SHT-AUC	UB-AUC
–	–	–	LVLM-only VAD	69.2%	–
✓	–	–	–	81.4%	69.0%
✓	✓	–	VE training first	81.9%	68.8%
–	✓	–	–	79.7%	64.5%
–	✓	✓	–	84.5%	71.6%
✓	✓	✓	**VE training first (default)**	**87.0%**	**73.4%**
✓	✓	✓	TE training first	86.3%	72.2%
✓	✓	–	Joint training	54.1%	56.7%
✓	✓	✓	Joint training	81.6%	67.7%

shown as red dots in Fig. 5. These experiments indicate that the CLIP pre-trained model exhibits strong generalization, effectively transferring features from synthetic anomalous samples to a broad spectrum of anomaly types.

4.5. Ablation study

We conducted ablation studies on ShanghaiTech and UBnormal datasets to assess the impact of each component within the SeS-CLIP model. This involved decomposing the model to evaluate the contributions of individual modules. To establish a baseline, we sequentially removed the synthetic anomalous sample step and the entire training phase of the text encoder. Subsequently, each module was gradually reintroduced to assess its specific contribution.

4.5.1. Effectiveness of the proposed components

The baseline model retains only the first training phase, representing the fundamental semi-supervised training paradigm without synthetic anomalous texts. As shown in Table 3, the baseline AUCs on ShanghaiTech and UBnormal are 81.4% and 69.0%, respectively, outperforming the majority of existing semi-supervised models. This emphasizes the advantages of integrating textual features and the pre-trained CLIP model into semi-supervised VAD frameworks. Fine-tuning the text encoder yielded negligible performance improvements relative to the baseline when synthetic anomalous text was not used, as expected, since image and text features are already aligned during the first training phase. The comparison across rows 3–6 of the table indicates that the significant improvements in our SeS-CLIP model are primarily due to the incorporation of synthetic anomalous texts. We further analyze the importance of our staged training strategy. As shown in row 8 of Table 3, joint training without synthetic anomalous texts causes severe performance degradation, suggesting substantial interference between the two optimization objectives, thereby supporting the effectiveness of the staged training strategy.

To evaluate the necessity of the entire Feature Contrast Module as an anomaly discriminator, we designed an experiment in which a single LVLM directly outputs a frame-level anomaly judgment in a single forward pass. By supplementing the prompt words from Section 3.2 with clear anomaly definitions, the LVLM can directly evaluate anomalies in each frame. As indicated in the first row of Table 3, the results were suboptimal, demonstrating the limitations of using a single Qwen2.5-VL-7B-Instruct model for video anomaly detection. To our knowledge, there is currently no competitive VAD method that relies solely on a frozen LVLM with approximately 10B parameters to process each frame in a single pass and directly produce frame-level anomaly results. Thus, achieving reliable outcomes still requires external discriminators for anomaly judgment or iterative multi-query interactions with the LVLM itself.

4.5.2. Effectiveness of the teacher vision encoder

We introduce several extreme anomaly examples to illustrate the necessity of incorporating a teacher vision encoder. While the inclusion of the teacher model yields a marginal drop in AUC on the standard

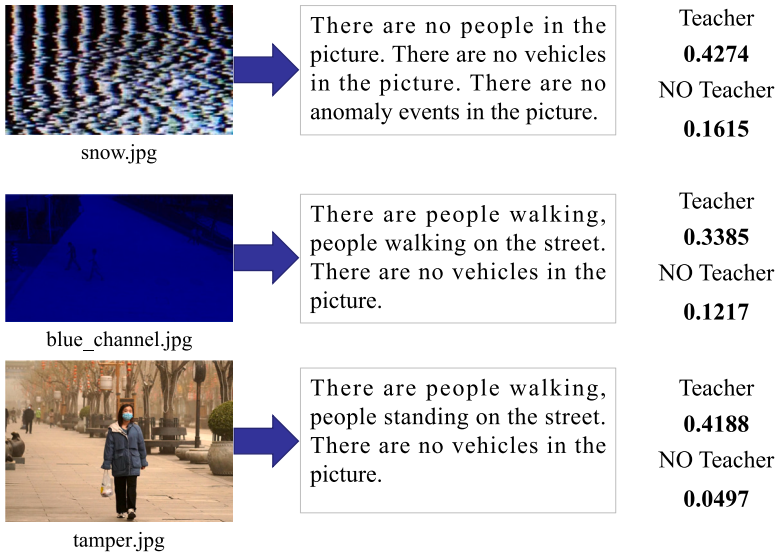


Fig. 7. Anomaly score comparison under the special anomaly types: with vs. without teacher vision encoder.

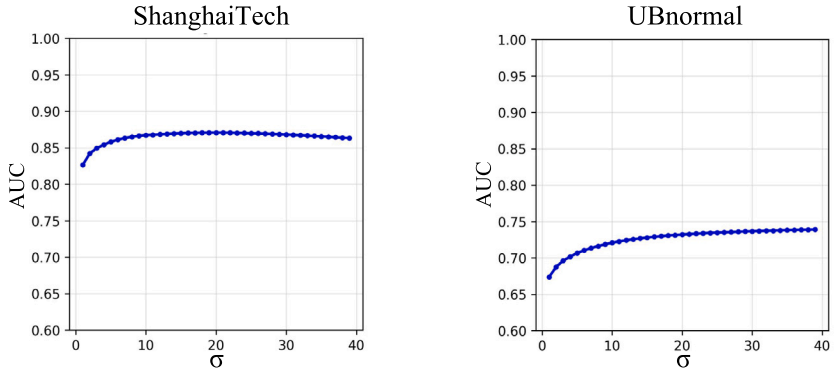


Fig. 8. Sensitivity to Gaussian filtering standard deviation σ. We smooth each video's frame-level anomaly score curve and evaluate AUC on the ShanghaiTech and UBnormal datasets.

Table 4
Maximum similarity between each abnormal image and its nearest normal sample in the test set.

Image	Without teacher model	With teacher model
snow.jpg	0.8249	0.2104
blue_channel.jpg	0.9076	0.7087
tamper.jpg	0.9215	0.2463

ShanghaiTech test set from 87.1% to 87.0%, its benefits become evident in out-of-distribution scenarios. As illustrated in Fig. 7 and Table 4, for certain extreme anomalous scenarios absent from the test set, such as camera damage or video tampering, the image background changes significantly. The LVLM may generate plausible descriptions like "There are people walking in the picture". In such cases, the model with a teacher vision encoder yields substantially higher anomaly scores than the one trained without it. These illustrative cases suggest that teacher-based feature regularization helps preserve general visual representations when the LVLM produces plausible but uninformative captions.

4.5.3. Effectiveness of the Gaussian filtering

We evaluated the AUC performance on the ShanghaiTech and UBnormal test sets under varying levels of Gaussian filtering, controlled by the standard deviation σ. According to Fig. 8, Gaussian filtering is necessary. However, when $\sigma > 7$, our model's performance reaches state-of-the-art (SOTA), and the AUC value becomes largely insensitive to further increases in σ. The performance remains relatively stable over a broad range of σ, indicating limited sensitivity to the exact smoothing strength. We further analyze this phenomenon through a representative scenario as shown in Fig. 9. Our model's detection failures are typically caused by occasional recognition errors made by the LVLM, such as mistaking walking for running or skateboarding, which result in high anomaly scores, or ignoring bicycles due to edge-of-image occlusion, which result in low anomaly scores. Compared to traditional generative VAD models, the resulting anomaly-score sequences may contain isolated temporal spikes caused by occasional LVLM recognition errors. In VAD tasks, we typically posit that anomalies occur within continuous video segments rather than scattered video frames, so adding temporal filtering can suppress isolated score fluctuations caused by occasional recognition errors.

4.5.4. Effectiveness of the LVLMs

To investigate the impact of different LVLMs on video anomaly detection performance, we replaced the original LVLM in our framework with other LVLMs and conducted a comparative evaluation. As shown in Table 5, LLaVA-v1.6-Vicuna-13B achieves performance comparable to that of Qwen2.5-VL-7B-Instruct. All evaluated LVLMs occasionally misdescribe normal walking or standing as anomalous activities such as skateboarding or frisbee throwing. Still, GEMMA3-27B-it has a misidentification rate of 8.36% in the ShanghaiTech training sets, much higher than Qwen2.5-VL-7B's 1.1% and LLaVA-v1.6-13B's

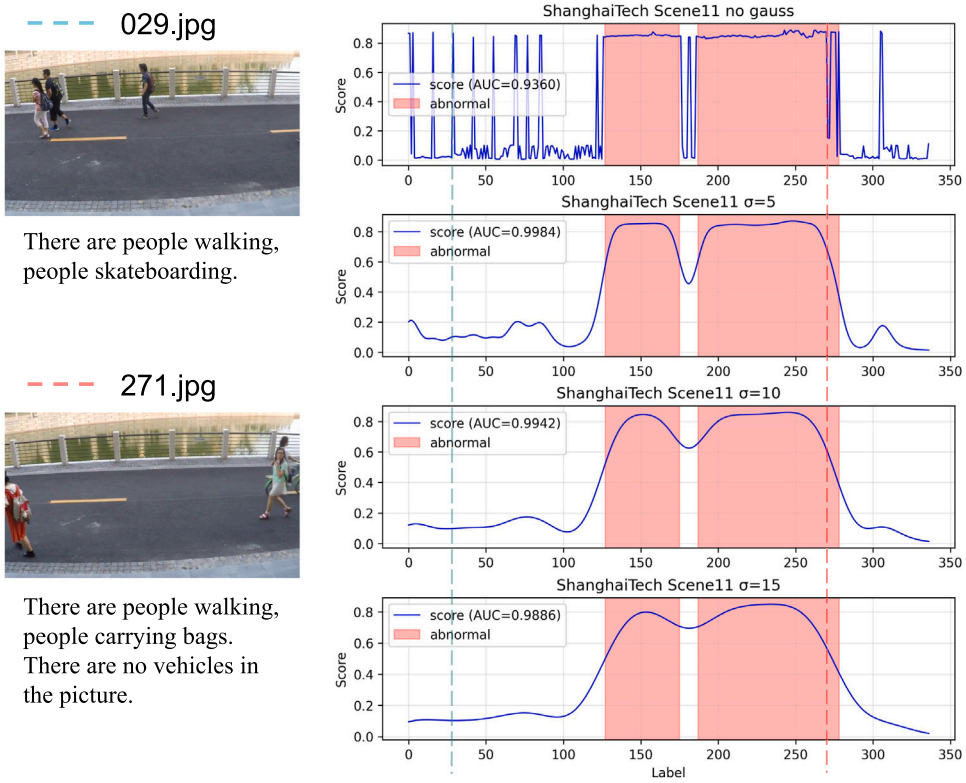


Fig. 9. Qualitative results by different standard deviations σ.

Table 5
AUC results with different LVLMs.

LVLM	SHT-AUC
Qwen2.5-VL-7B-Instruct (default)	87.0%
Qwen2.5-VL-3B-Instruct	84.5%
GEMMA3-27B-it	83.8%
LLaVA-v1.6-vicuna-13B	86.2%
Qwen3-VL-8B-Instruct	88.2%

Table 6
AUC results with different temporal window radii and top-K settings.

Temporal window radius (Frames)	Top-K	SHT-AUC	UB-AUC
0 (self-matching)	–	85.3%	71.7%
3	1	**87.2%**	72.3%
2	2	85.9%	72.8%
3	2	86.1%	73.0%
3 (default)	**3**	**87.0%**	**73.4%**
4	3	86.9%	**73.5%**
4	4	86.3%	72.7%

0.83%, which likely contributes to the lower VAD performance of GEMMA3-27B-it. In addition, we tested the performance of the newer Qwen3-VL-8B-Instruct model, and the score improved further to 88.2%, suggesting that SeS-CLIP may benefit from stronger LVLM backbones.

4.5.5. Effect of temporal neighborhood size and top-K selection

We further validate the design of L^{v1} from Eqs. (5)–(7) by varying the temporal window radius and the top-K selection. Table 6 reports results for a self-matching baseline and six temporal-neighborhood configurations. The default configuration with a radius of 3 and top-3 selection provides a favorable trade-off across the two datasets, achieving 87.0% AUC on ShanghaiTech and 73.4% on UBnormal. Notably, every variant outperforms the per-frame self-matching scheme, suggesting that frame-text matching on a short temporal pool of captions rather than self-matching reduces vulnerability to caption errors, thus alleviating harmful image-text mismatches.

Table 7
Average anomaly scores at different training stages on the ShanghaiTech dataset, where S_n and S_a denote the average scores for normal frames and anomalous frames, respectively.

	S_n	S_a
CLIP pretrained model	0.412	0.410
After vision encoder training	0.103	0.208
After text & vision encoder training	0.133	0.587

4.5.6. Effect of synthetic anomaly text count

As shown in Fig. 10, we vary the number of synthetic anomaly samples per batch and report the resulting AUC on the ShanghaiTech and UBnormal test sets. Introducing synthetic samples yields a substantial AUC improvement over the baseline that uses no synthetic samples; this trend is also borne out quantitatively in Table 3. However, beyond 9 synthetic samples per batch, the marginal gain diminishes sharply, and the performance essentially plateaus.

4.6. Qualitative results

We illustrate the performance of the anomaly score across various types of anomalous videos. We first examine the overall distribution of anomaly scores on the ShanghaiTech dataset. As shown in Table 7, the CLIP pretrained model yields an average score of 0.412 over normal frames and 0.410 over anomalous frames. After the first training stage, the average scores shift to 0.103 for normal frames and 0.208 for anomalous frames. Following the second training stage, the gap between normal and anomalous frames further widens, with average scores of 0.133 and 0.587, respectively, which is in line with our expectations in Fig. 4.

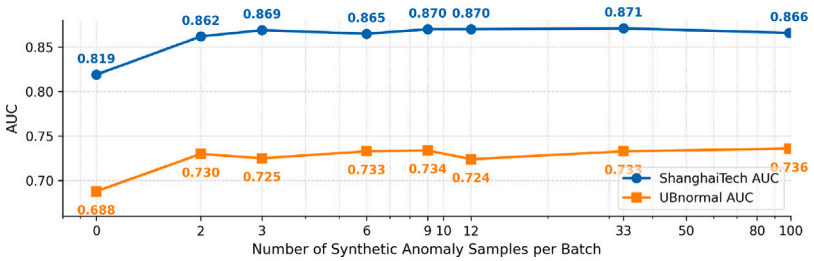


Fig. 10. AUC score under different numbers of synthetic anomalous captions.

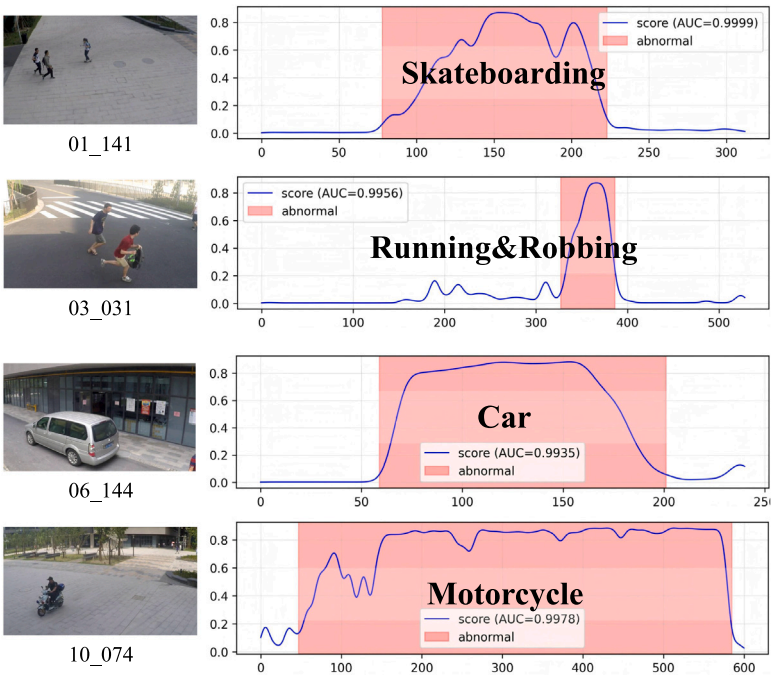


Fig. 11. Anomaly score curves of several test video samples on the ShanghaiTech dataset.

Table 8
AUC results under different downsampling factors.

Model	γ	SHT-AUC
Qwen2.5-VL-7B	1	87.0%
	2	86.6%
	3	87.2%
	4	86.1%
	5	87.1%
	6	86.4%
	7	84.7%

As shown in Fig. 11, when different anomalous events occur, the score exhibits a sharp increase, and upon the conclusion of the event, it promptly returns to a lower range. For normal frames, our method consistently maintains the anomaly score within a very low range. The score curves provide clear evidence of our method's high sensitivity to anomalous events, enabling accurate and timely detection while maintaining low anomaly-score predictions for normal samples.

4.7. Runtime and real-time performance

We analyzed the running speed of our model and the feasibility of real-time detection on a single NVIDIA RTX A6000 Graphics card. Due to the use of an LVLM, the speed of our structured text generation module is 4.6 frames per second (FPS) on Qwen2.5-VL-7B and 5.9 FPS on Qwen3-VL-8B. The feature contrast module runs at 82 FPS and is therefore not the primary computational bottleneck. To reduce computational cost and meet the requirement for 24 FPS real-time processing, potential strategies include employing quantized models, utilizing more advanced LVLM or GPU hardware, or reducing the video frame rate to $1/\gamma$ of its original value to reduce data volume. As shown in Table 8, due to the inherent temporal continuity of video sequences, downsampling at rates below 6-fold has a negligible impact on AUC, indicating that our model can run in real time.

5. Conclusions

This paper presents a novel semi-supervised framework for VAD. By leveraging an LVLM to generate structured text descriptions for video frames, our approach enables the CLIP-based feature contrast module to learn the feature distribution of normal image-text pairs effectively. This method enhances the capture of high-level semantic information from text features, surpassing the capabilities of other semi-supervised models. Furthermore, by pioneering the integration of synthetic anomalous texts, we amplify the feature disparities between image-text pairs in anomalous samples. Extensive experiments demonstrate the superiority of our approach, confirming its effectiveness over existing semi-supervised models.

CRediT authorship contribution statement

Shiyu Zhang: Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis,

Data curation, Conceptualization. **Cheng Yan:** Writing – review & editing, Validation, Methodology, Investigation. **Wenjun Wang:** Writing – review & editing, Resources, Funding acquisition. **Lei Zhou:** Methodology.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Cheng Yan reports financial support was provided by National Key R&D Program of China (No. 2023YFC3805203). Wenjun Wang reports financial support was provided by National Natural Science Foundation of China (No. 62472305). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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